

HDOC Day-8

Question 1

Write a C program to find the surface area and volume of a cylinder in square meters and cubic meters when radius and height are given in centimeters.

Step 1: Understanding the Problem

Input: Radius r in centimeters (cm) and height h in centimeters (cm).

Output: Total surface area in square meters (m^2) and volume in cubic meters (m^3).

Formula(s):

- Convert: $r(m) = r(cm) / 100$ and $h(m) = h(cm) / 100$
- Total surface area = $2 * PI * r * (r + h)$
- Volume = $PI * r^2 * h$

Step 2: Standard Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result (Pass/Fail)
1	$r=10$ cm, $h=20$ cm	Area= $0.188496 m^2$; Volume= $0.006283 m^3$	To be filled after execution	Pass
2	$r=5$ cm, $h=10$ cm	Area= $0.047124 m^2$; Volume= $0.000785 m^3$	To be filled after execution	Pass
3	$r=25$ cm, $h=50$ cm	Area= $1.178097 m^2$; Volume= $0.098175 m^3$	To be filled after execution	Pass

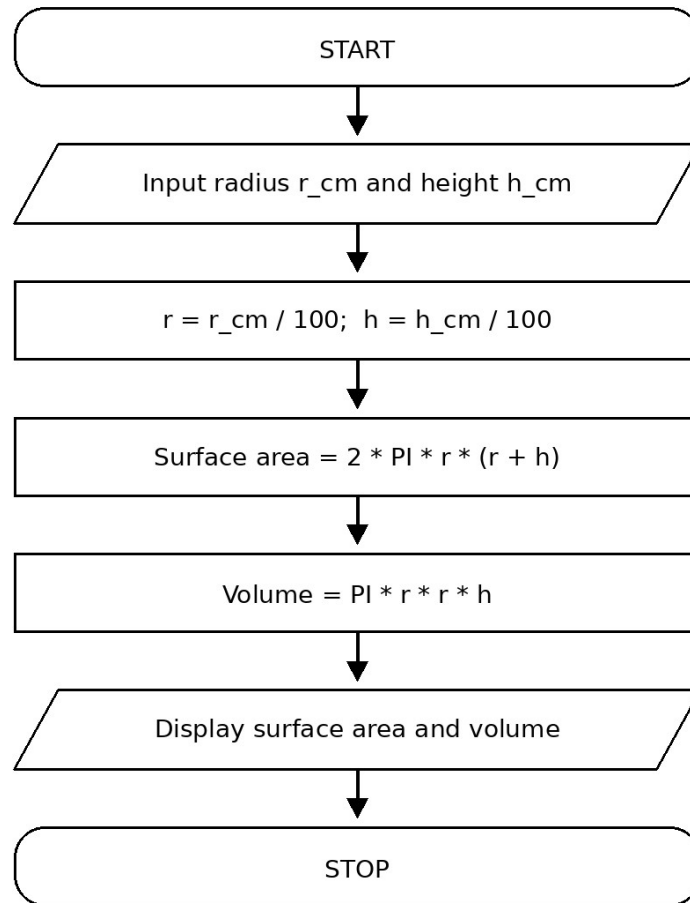
Note: After running the program in Kali Linux, compare your displayed values with the Expected Output column. If they match to the shown decimal places, write Pass.

Step 3: Algorithm and Evaluation

1. Start.
2. Read radius in cm and height in cm.
3. Convert radius and height from cm to m by dividing each by 100.
4. Calculate total surface area using $2 * PI * r * (r + h)$.
5. Calculate volume using $PI * r * r * h$.
6. Display surface area in m^2 and volume in m^3 .
7. Stop.

Algorithm evaluation: For $r=10$ cm and $h=20$ cm, conversion gives 0.10 m and 0.20 m; the formulas give $0.188496 m^2$ and $0.006283 m^3$. The other two test cases are checked in the same way.

Step 4: Flowchart and Evaluation



Flowchart evaluation: Follow the flowchart using each test-case input. The calculations lead to the same expected outputs listed in Step 2.

Step 5: C Program

```
#include <stdio.h>

int main()
{
    double radius_cm, height_cm;
    double radius_m, height_m;
    double surface_area, volume;
    const double PI = 3.141592653589793;

    printf("Enter radius in centimeters: ");
    scanf("%lf", &radius_cm);
    printf("Enter height in centimeters: ");
    scanf("%lf", &height_cm);

    radius_m = radius_cm / 100.0;
    height_m = height_cm / 100.0;

    surface_area = 2.0 * PI * radius_m * (radius_m + height_m);
    volume = PI * radius_m * radius_m * height_m;
```

```
printf("Surface Area = %.6f square meters\n", surface_area);  
printf("Volume = %.6f cubic meters\n", volume);  
  
return 0;  
}
```

Step 6: Compile, Execute and Evaluate in Kali Linux

```
(kali㉿kali)-[~/Desktop]  
$ vi question1.c  
  
(kali㉿kali)-[~/Desktop]  
$ gcc question1.c -o question1  
  
(kali㉿kali)-[~/Desktop]  
$ ./question1  
Enter radius in centimeters: 10  
Enter height in centimeters: 20  
Surface Area = 0.188496 square meters  
Volume = 0.006283 cubic meters  
  
(kali㉿kali)-[~/Desktop]  
$ ./question1  
Enter radius in centimeters: 5  
Enter height in centimeters: 10  
Surface Area = 0.047124 square meters  
Volume = 0.000785 cubic meters  
  
(kali㉿kali)-[~/Desktop]  
$ ./question1  
Enter radius in centimeters: 25  
Enter height in centimeters: 50  
Surface Area = 1.178097 square meters  
Volume = 0.098175 cubic meters
```

Evaluation: Run all three test cases. Compare the terminal output with Step 2. When the values match, mark the corresponding result as Pass.

Step 7: Compilation and Execution Screenshots

```
(kali㉿kali)-[~/Desktop]  
$ vi question1.c  
  
(kali㉿kali)-[~/Desktop]  
$ gcc question1.c -o question1
```

```
(kali㉿kali)-[~/Desktop]  
$ ./question1  
Enter radius in centimeters: 10  
Enter height in centimeters: 20  
Surface Area = 0.188496 square meters  
Volume = 0.006283 cubic meters
```

```
(kali㉿kali)-[~/Desktop]
$ ./question1
Enter radius in centimeters: 5
Enter height in centimeters: 10
Surface Area = 0.047124 square meters
Volume = 0.000785 cubic meters
```

```
(kali㉿kali)-[~/Desktop]
$ ./question1
Enter radius in centimeters: 25
Enter height in centimeters: 50
Surface Area = 1.178097 square meters
Volume = 0.098175 cubic meters
```

Question 2

Write a C program to find the area and perimeter of a semicircle when radius is given in millimeters and output is required in centimeters.

Step 1: Understanding the Problem

Input: Radius r in millimeters (mm).

Output: Area in square centimeters (cm^2) and perimeter in centimeters (cm).

Formula(s):

- Convert: $r(\text{cm}) = r(\text{mm}) / 10$
- Area of semicircle = $(\text{PI} * r^2) / 2$
- Perimeter of semicircle = $\text{PI} * r + 2r$

Step 2: Standard Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result (Pass/Fail)
1	$r=10$ mm	Area= 1.5708 cm^2 ; Perimeter= 5.1416 cm	To be filled after execution	Pass
2	$r=20$ mm	Area= 6.2832 cm^2 ; Perimeter= 10.2832 cm	To be filled after execution	Pass
3	$r=50$ mm	Area= 39.2699 cm^2 ; Perimeter= 25.7080 cm	To be filled after execution	Pass

Note: After running the program in Kali Linux, compare your displayed values with the Expected Output column. If they match to the shown decimal places, write Pass.

Step 3: Algorithm and Evaluation

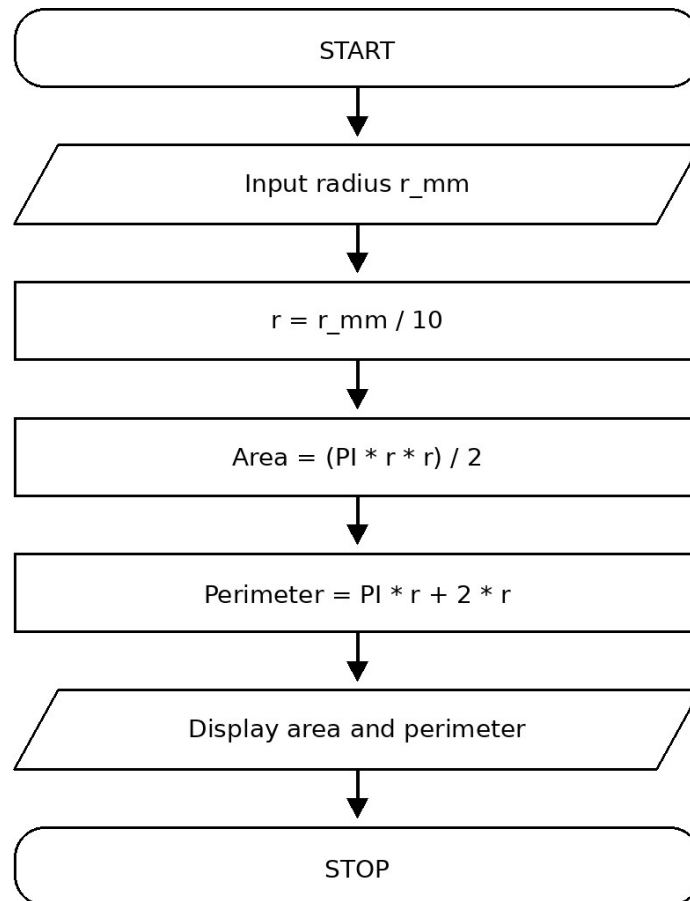
1. Start.
2. Read radius in mm.
3. Convert radius from mm to cm by dividing by 10.
4. Calculate area using $(\text{PI} * r * r) / 2$.
5. Calculate perimeter using $\text{PI} * r + 2 * r$.

6. Display area in cm^2 and perimeter in cm.

7. Stop.

Algorithm evaluation: For $r=10$ mm, conversion gives $r=1$ cm; the formulas give area 1.5708 cm^2 and perimeter 5.1416 cm. The remaining test cases are evaluated by the same steps.

Step 4: Flowchart and Evaluation



Flowchart evaluation: Follow the flowchart using each test-case input. The calculations lead to the same expected outputs listed in Step 2.

Step 5: C Program

```
#include <stdio.h>

int main()
{
    double radius_mm, radius_cm;
    double area, perimeter;
    const double PI = 3.141592653589793;

    printf("Enter radius in millimeters: ");
    scanf("%lf", &radius_mm);
```

```
radius_cm = radius_mm / 10.0;

area = (PI * radius_cm * radius_cm) / 2.0;
perimeter = PI * radius_cm + 2.0 * radius_cm;

printf("Area = %.4f square centimeters\n", area);
printf("Perimeter = %.4f centimeters\n", perimeter);

return 0;
}
```

Step 6: Compile, Execute and Evaluate in Kali Linux

```
(kali㉿kali)-[~/Desktop]
$ vi question2.c

(kali㉿kali)-[~/Desktop]
$ gcc question2.c -o question2

(kali㉿kali)-[~/Desktop]
$ ./question2
Enter radius in millimeters: 10
Area = 1.5708 square centimeters
Perimeter = 5.1416 centimeters

(kali㉿kali)-[~/Desktop]
$ 20
20: command not found

(kali㉿kali)-[~/Desktop]
$ ./question2
Enter radius in millimeters: 20
Area = 6.2832 square centimeters
Perimeter = 10.2832 centimeters

(kali㉿kali)-[~/Desktop]
$ ./question2
Enter radius in millimeters: 50
Area = 39.2699 square centimeters
Perimeter = 25.7080 centimeters
```

Evaluation: Run all three test cases. Compare the terminal output with Step 2. When the values match, mark the corresponding result as Pass.

Step 7: Compilation and Execution Screenshots

```
(kali㉿kali)-[~/Desktop]
$ vi question2.c

(kali㉿kali)-[~/Desktop]
$ gcc question2.c -o question2
```

```
(kali㉿kali)-[~/Desktop]  
$ ./question2  
Enter radius in millimeters: 10  
Area = 1.5708 square centimeters  
Perimeter = 5.1416 centimeters
```

```
(kali㉿kali)-[~/Desktop]  
$ ./question2  
Enter radius in millimeters: 20  
Area = 6.2832 square centimeters  
Perimeter = 10.2832 centimeters
```

```
(kali㉿kali)-[~/Desktop]  
$ ./question2  
Enter radius in millimeters: 50  
Area = 39.2699 square centimeters  
Perimeter = 25.7080 centimeters
```